# 🤖 Dialogflow Chatbot System using Dialogflow ES, .NET Core, Angular & SignalR

## 📌 Project Overview

This project is a real-time chatbot system that communicates with users through a web interface. It integrates **Dialogflow ES** for natural language understanding and uses **SignalR (WebSockets)** for real-time communication.

The chatbot can:

- Answer predefined intents
- Retrieve answers from a Knowledge Base (FAQ)
- Maintain chat history per session
- Provide real-time responses without page refresh

## 🏗 Architecture

```
User (Angular UI)
        ↓
SignalR (WebSocket Connection)
        ↓
ASP.NET Core Backend (Hub + API)
        ↓
Dialogflow ES (Intent + Knowledge Base)
        ↓
Response back to User (real-time)
```

# 🛠 Technologies Used

## Frontend

- Angular
- TypeScript
- Bootstrap (UI styling)

## Backend

- ASP.NET Core Web API
- SignalR (real-time communication)

## AI/NLP

- Dialogflow ES (Google Cloud)

## Other

- C#
- HTML/CSS
- LocalStorage (Session handling)

# ⚙ Key Features

## 1. Real-time Communication

- Implemented using SignalR
- No need to refresh UI
- Instant message delivery

## 2. Session Management

- Unique `sessionId` generated
- Stored in browser `localStorage`
- Maintains chat history per user

### 3. Chat Memory (Without Database)

- Messages stored in backend using list
- Each session has its own message list

### 4. Other

- Basic Error Handling
- Auto Scroller
- Message UI Design
- Typing Indicator

## 🔄 Chat Flow

1. User types message
2. Message sent via SignalR
3. Backend:
    a. Stores message
    b. Sends to Dialogflow
4. Dialogflow:
    a. Matches Intent OR Knowledge Base
5. Response returned to backend
6. Backend sends response via SignalR
7. UI updates instantly

## ⚠️ Limitations

- No database (data lost on server restart)
- Basic NLP (Dialogflow ES limitations)

## 🚀 Future Enhancements

- Add database
- Implement user authentication
- Add more Chat Options

## 📌 Conclusion

This project demonstrates how to build a **real-time AI chatbot** using modern technologies. It combines:

- Real-time communication (SignalR)
- AI/NLP (Dialogflow)
- Scalable backend (.NET Core)
- Interactive frontend (Angular)

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